MIL-STD-17B-2
23 January 1963
SUPERSEDING
(See FOREWORD)

MILITARY STANDARD

MECHANICAL SYMBOLS FOR AERONAUTICAL, AEROSPACECRAFT AND SPACECRAFT USE

PART 2

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DEPARTMENT OF DEFENSE

WASHINGTON 25, D. C.

Mechanical Symbols for Aeronautical,
Aerospacecraft and Spacecraft Use

MIL-STD-17B-2

1. This standard has been approved by the Department of Defense and is mandatory for use by the Department of the Army, the Navy, and the Air Force, effective 23 January 1963.

2. Recommended corrections, additions, or deletions should be addressed to Standardization Division, Defense Supply Agency, Alexandria, Virginia.

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FOREWORD

MIL-STD-17B-2 contains a listing of graphical mechanical drawing symbols for aeronautical, aerospacecraft and spacecraft use. These mechanical symbols should be used by all who have occasion to prepare schematic drawings for aeronautical, aerospacecraft and spacecraft for the Departments of the Army, the Navy, and the Air Force.

The symbols formerly contained in the basic issue of Standard MIL-STD-17, dated 6 July 1950, section 3 were considered carefully in the preparation of Standard MIL-STD-17B-2 and where there was no good reason for making a change, their use was adopted. Certain new symbols have been added.

Standard MIL-STD-17A has been revised and published as MIL-STD-17B-1. MIL-STD-17 now consists of the following two parts:

Part 1 - Mechanical Symbols (Other Than Aeronautical, Aerospacecraft and Spacecraft Use)

Part 2 - Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft Use

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MECHANICAL SYMBOLS FOR

AERONAUTICAL, AEROSPACECRAFT AND SPACECRAFT USE

1. SCOPE

1.1 This standard establishes an approved list of graphical symbols that shall be used in preparation of schematic diagrams for fluid, gaseous and vacuum systems in aeronautical, aerospacecraft and spacecraft equipments diagramatically.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal forms a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-25516 - Connectors, Electrical, Miniature, Coaxial, Shielded and Unshielded, Environment Resistant Type, General Specification For

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. DEFINITIONS (Not Applicable)

4. SYMBOLS

4.1 Tube and Hose Lines

4.1.1 Power plant, instrument, fire extinguisher and miscellaneous tube and hose line symbols.

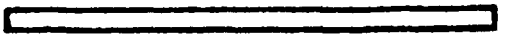
Service	Riser or stack	Symbol
Bellows		
Bleed return	(BR)	
Drain	(D)	
Engine coolant	(EC)	
Foam solution	(FO)	
Fuel oil vent	(FOV)	
Fuel supply	(F)	

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Symbol	Riser or stack	Service
——— LO ———	LO	Lubricating oil
— — — OB — — —	OB	Oil breather
——— OD ———	OD	Oil dilution
——— PN ———	PN	Pneumatic
		Self-sealing
——— V ———		Vacuum
— — — — V — — — —	V	Vent
— — — PV — — —	PV	Vent pressure
— — — — VR — — — —	VR	Vent return
——— WI ———	WI	Water injection
——— - ——— - ———		Water, cold
——— .. ——— .. ———		Water, hot

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4.1.2 Hydraulic tube and hose line symbols

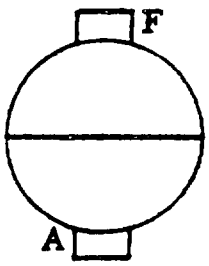
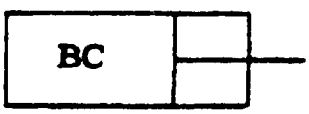
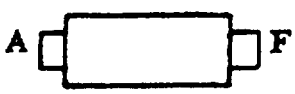
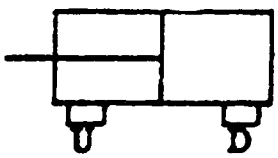
Service	Symbol
Brake	
Down (or close)	
Emergency pressure	
Hose connection, rigid tubing	
Hose, flexible	
Return	
Supply fluid, pump suction	
Suction gravity	
Supply pressure	
Up (or open)	
Vent	

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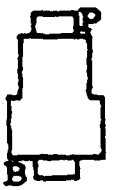
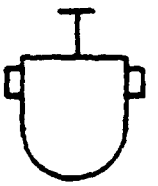
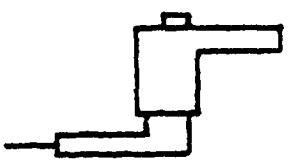
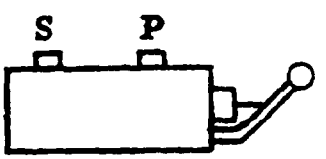
4.2 HYDRAULIC EQUIPMENT

4.2.1 General. The hydraulic symbols shown merely present outlines. The schematic diagram should show the cutaway section of each unit at least in schematic form.

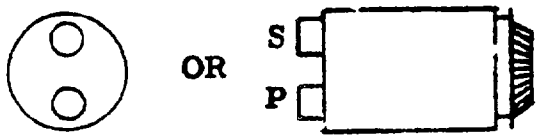
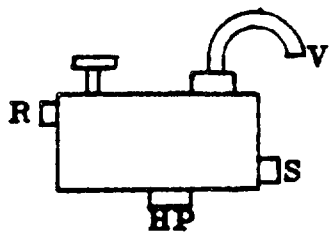
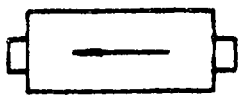
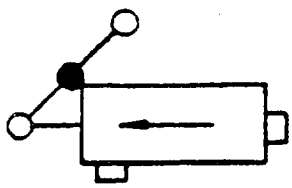
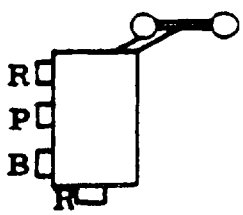
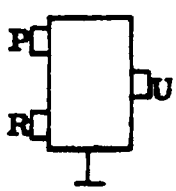
4.2.2 Hydraulic equipment symbols

Symbol	Service
	Accumulator
	Air bottle, emergency
	Brake control
	Bungee, air-oil
	Coupling, self-sealing
	Cylinder, actuating

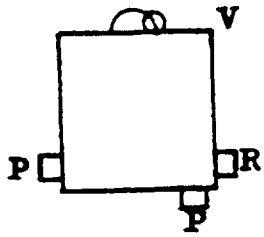
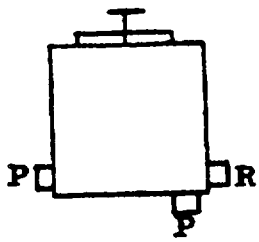
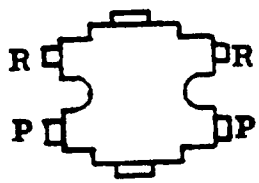
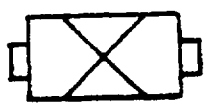
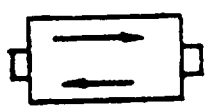
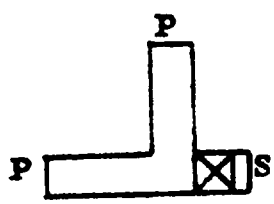
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Service	Symbol
Debooster, brake	
Filter or strainer	
Fitting, swivel	
Gage, pressure	
Gage and snubber, pressure	
Pump, hand	

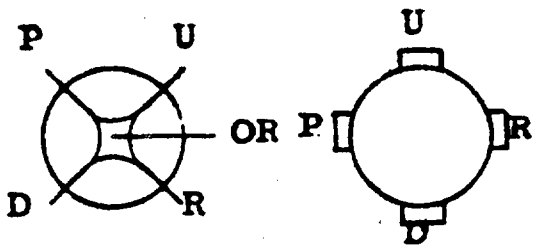
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Symbol	Service
	Pump, power driven
	Reservoir
	Valve, check, automatic
	Valve, check, manual
	Valve, brake control
	Valve, gun charger control

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Service	Symbol
Valve, pressure regulating (unloading) automatic	
Valve, pressure regulating (unloading) manual	
Valve, relief	
Valve, restrictor, both ways	
Valve, restrictor, partial one-way	
Valve, shuttle	

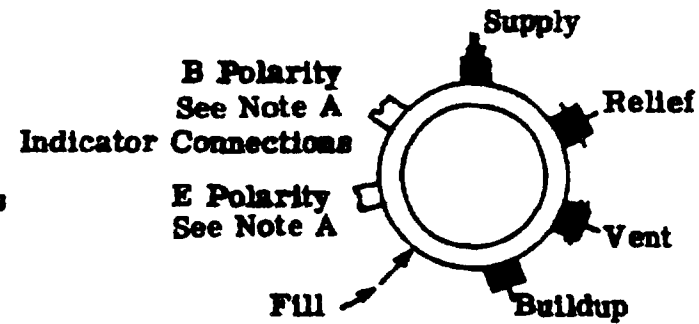
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Symbol	Service
	Valve or selector, directional control

4.3 OXYGEN EQUIPMENT

4.3.1 Tube and Hose Line Symbols

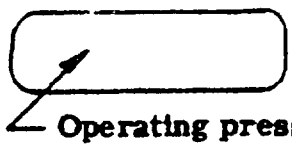
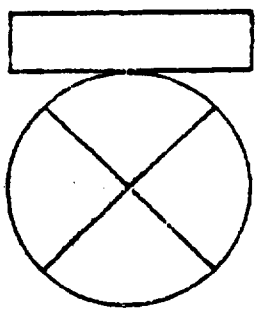
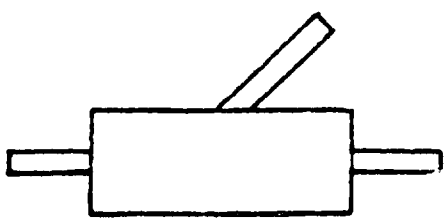
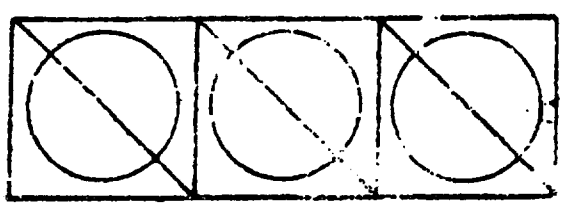
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Service	Symbol
Liquid oxygen converter with connection symbols	
Distribution Line, High pressure (2000 psi)	
Distribution line, low pressure (70 to 500 psi)	
Filler line	
Electrical line	
Flexible line	
Warming equipment	
Overboard vent line	

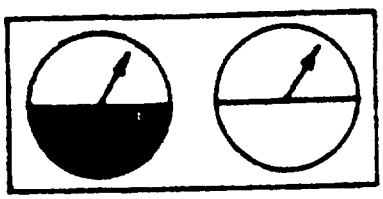
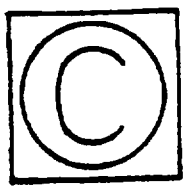
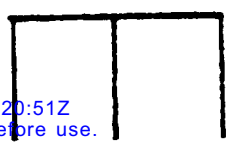
NOTE A - See MIL-C-25516

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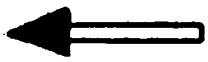
4.3.2 Oxygen Equipment Symbols

Symbol	Service
	Mask mounted regulator
	Recharger assembly
	Oxygen cylinder
	Aneroid operated valve
	On - off valve (toggle)
	Automatic mask presents (3 unit)

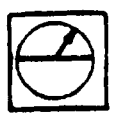
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Service	Symbol
Liquid oxygen drain valve	
Quantity indicator	
Repeater quantity indicator	
Oxygen control panel	
Pressure Suit Controller	
Pressure gage	
Automatic coupling (disconnect)	

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Symbol	Service
	Check valve, style A
	Check valve, style B
	Check valve, style C
	Check valve, style D
	Check valve, style E
	Check valve, style F
	Check valve, style G
	Check valve, style H
	Check valve, spring loaded
	13 Fill valve

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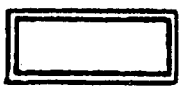
Service	Symbol
Portable unit (continuous flow)	
Portable unit (demand)	
Portable unit (pressure demand)	
Pressure reducer	
Automatic, continuous flow regulator	
Diluter demand regulator	
Helmet regulator	
Seat kit regulator	
Mask - regulator hose	
Helmet - regulator hose	

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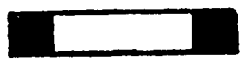
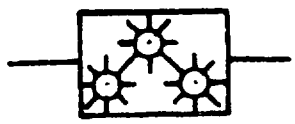
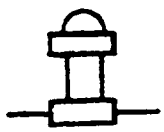
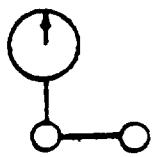
Symbol	Service
	Line valve
	Fill - buildup - vent valve

4.4 POWER PLANT EQUIPMENT

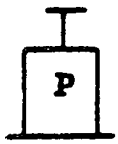
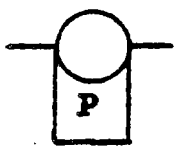
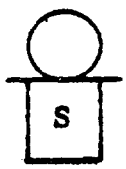
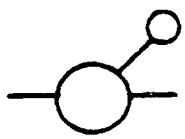
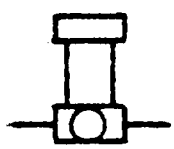
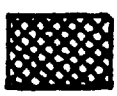
4.4.1 Power Plant Equipment Symbols

Symbol	Service
	Cooler, oil
	Detector, carbon monoxide
	Filler neck
	Filter, fuel
	Filter, induction air

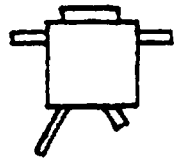
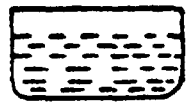
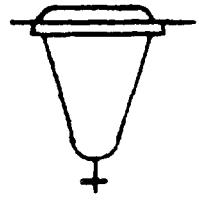
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Service	Symbol
Flowmeter	
Gage glass	
Gear box	
Governor, propeller	
Indicator, quantity	
Motor, pneumatic	
Pump	
Pump, air	
Pump, auxiliary fuel, electric driven	

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Symbol	Service
	Pump, coolant
	Pump, primer, hand
	Pump, primer, motor driven
	Pump, scavenger
	Pump, wobble
	Pump and motor, propeller feather
	Radiator, coolant
	Scupper

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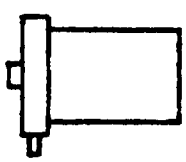
Service	Symbol
Separator, oil	
Strainer	
Sump	
Tank	
Tank, droppable	
Tank, expansion	
Tank, self-sealing	
Trap, water	

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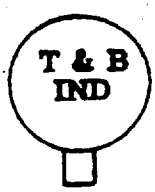
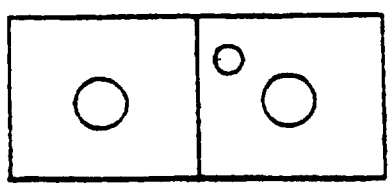
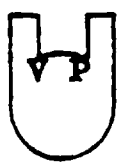
Symbol	Service
	Valve, defueling
 2 WAY 3 WAY 4 WAY	Valve, fuel selector

4.5 VACUUM INSTRUMENT EQUIPMENT

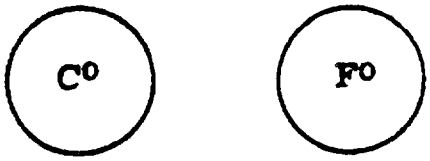
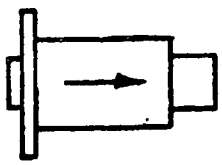
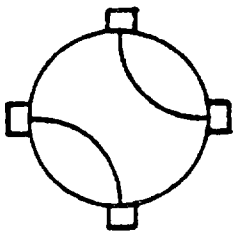
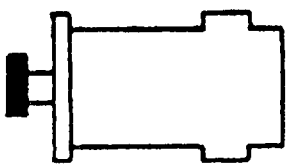
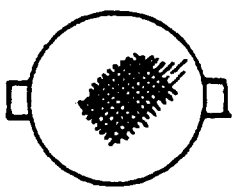
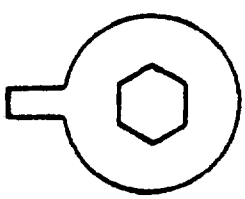
4.5.1 Vacuum instrument equipment symbols

Symbol	Service
	Air filter - turn and bank indicator
	Boot, de-icer
	Filter, air
	Gage, vacuum

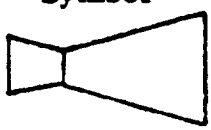
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Service	Symbol
Indicator, directional gyro	
Indicator, gyro horizon	
Indicator, turn and bank	
Outlet, vacuum	
Pilot, automatic	
Pump, vacuum	
Selector, static pressure	

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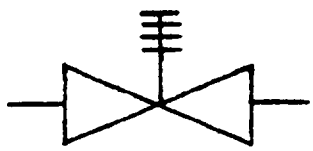
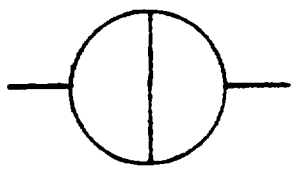
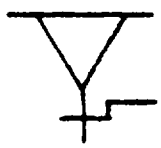
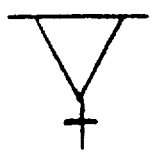
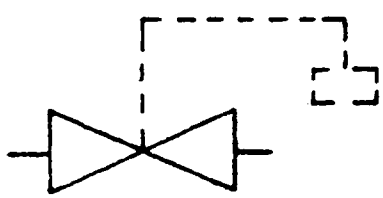
Symbol	Service
	Thermometer
	Valve, air check
	Valve, selector, four-way
	Valve, vacuum control, turn and bank indicator
	Valve, vacuum relief
	Valve, vacuum throttling

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Check the source to verify that this is the current version before use.

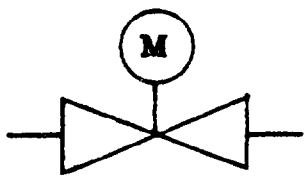
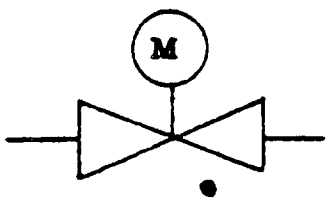
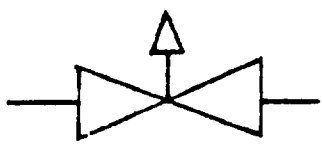
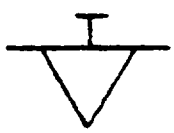
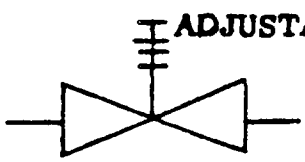
Service	Symbol
Venturi tube	

4 6 MISCELLANEOUS VALVES

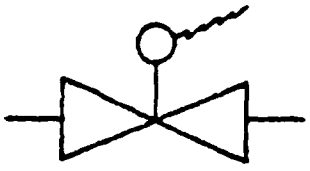
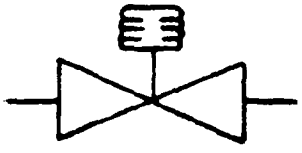
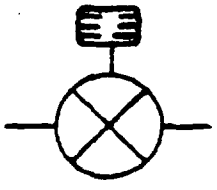
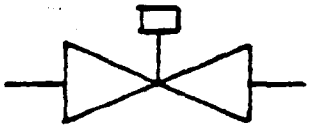
4.6 1 Miscellaneous valve symbols

Service	Symbol
Back pressure	
Cross feed	
Dump	
Drain	
Float operated	

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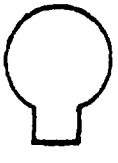
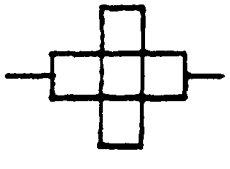
Symbol	Service
	Gate
	Gate, motor operated
	Globe
	Globe, motor operated
	Needle
	Poppet
	Pressure regulator

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Check the source to verify that this is the current version before use.

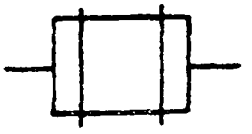
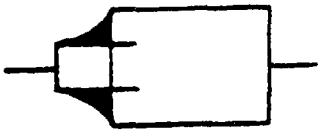
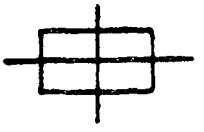
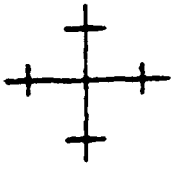
Service	Symbol
Solenoid	
Supercharger regulator	
Temperature control	
Water regulator	

4.7 FITTINGS

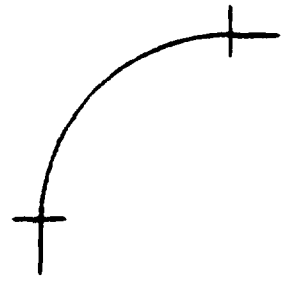
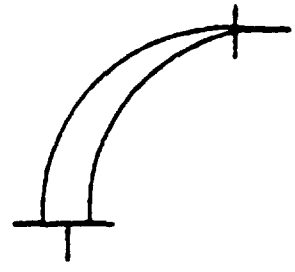
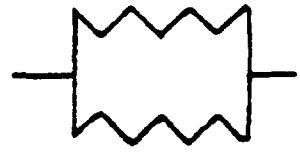
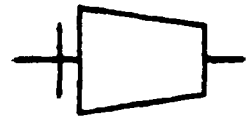
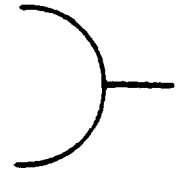
4.7.1 Fitting Symbols

Service	Symbol
Banjo	
Bulkhead fitting	

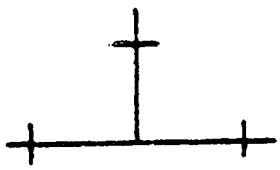
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Symbol	Service
	Bushing
	Cap
	Connector, hose
	Connector, nut and sleeve
	Coupling, female
	Cross
	Elbow, 45 degrees

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Service	Symbol
Elbow, 90 degrees	
Elbow, reducing	
Expansion joint, bellows	
Expansion joint, sliding	
Flange, reducing	
Nipple	

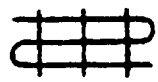
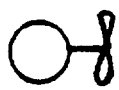
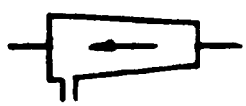
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Symbol	Service
	Reducer, concentric
	Tee
	Union, flanged
	Union, screwed

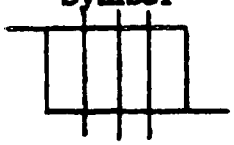
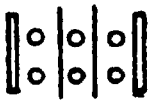
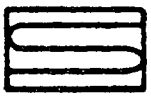
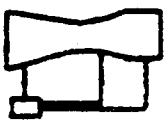
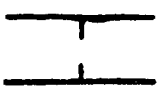
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4.8 ENVIRONMENTAL CONTROL EQUIPMENT

4.8.1 ENVIRONMENTAL CONTROL EQUIPMENT SYMBOLS

Service	Symbol
Air cycle cooling unit	
Heat exchanger, air to air	
Heat exchanger, liquid to liquid	
Heat exchanger, liquid to air	
Pump, liquid	
Heater, electric	
Heater, combustion	
Blower	
Fan	
Water separator	
Ejector, or jet pump	

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Symbol	Service
	Condenser, refrigerant
	Compressor, refrigerant
	Evaporator, refrigerant
	Radiator, to space
	Check valve
	Flow control valve
	Shut-off-valve
	Modulating valve
	Pressure regulating valve
	Relief valve
	Orifice, flow control
	Temperature controller
	Storage tank
	Cryogenic storage container

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